

TED (15)—6011 Reg. No: -----

(REVISION—2015)

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**DIPLOMA EXAMINATION IN ENGINEERING/ TECHNOLOGY/MANAGEMENT
/COMMERCIAL PRACTICE**

STRUCTURAL & IRRIGATION ENGINEERING DRAWING

MODEL QUESTION PAPER[Time : 3Hours]

(Maximum marks: 100)

[Note: 1) Missing data may be suitably assumed.

2) Steel tables are permitted

3) A2 size drawing sheet to be supplied.

4) Drawings shall be neat and fully dimensioned.

5) Answer one full question from each unit.

MODULE—I

I A simply supported two way slab has the following details :

Size of room –3.50 x4.50 m

Clear cover to Reinforcement 20mm

Main reinforcement along shorter span 10mm Φ @ 200mm c/c.

Main reinforcement along longer span 8mm Φ @ 150mm C/C

Thickness of slab 125mm

Draw (a) Plan showing bottom reinforcement. (10)

(b) Section along shorter span (15)

OR

II A singly reinforced beam has the following particulars.

Clear span of beam—5 m. Wall thickness.300mm

Cross section of beam --250 x450 mm. Clear cover to Stirrups—25mm.

Beam --tension reinforcement: 3 Nos of 16 mm Φ with one of them bend up at 0.15 le.

Stirrups -6mm Φ @ 100mm C/C up to 1m from centre of supportand remaining @ 125mm C /C.

Draw (1) Cross section of the beam at centre and support. (10)

(2) Longitudinal section (15)

MODULE—II

III The following are the details of a RCC retaining wall.

Natural G L+6.00. Top level of Retaining wall +10.500 Top level of base slab +5.50.

Bottom of base slab +.4.900 Toe projection 1000mm. Size of base slab 3500x600mm

Top width of stem - 200mm Bottom width of stem - 500mm

Stem Main steel-20mm Φ @ 120mm C/C. Alternate bars curtailed at 1/2height from top.

Distributors 10mm Φ @ 150mm C/C. Temperature steel: 10 mm Φ @ 150 mm C/C in both direction.

Toe and Heel main steel: 16mm Φ @ 150mm C/C. Distributors: 10mm Φ @ 150mm C/C.

Draw: 1) Longitudinal elevation of stem reinforcement with curtailment (10)

2) Cross Section of retaining wall and base slab (15)

OR

IV The details of a R.C C T-beam and slab bridge are given below.

Clear width of roadway 6.2m, footpath on either side 1.2 m wide, are provided.

Number of beams 2Nos Size of rib of beam 1400 x 450 mm. Thickness of slab 200 mm

Reinforcement: in slab 16 mm diameter rods at 150 mm C/C continuous over beams.

Distributors 12 mm Φ rods at 200 mm C/C.

Beam Reinforcement details: 5 Nos of 40 mm Φ rods provided in 2 rows.

Stirrups: 4 legged vertical stirrups 8mm Φ at 180mm C/C.

Footpath: Precast R.C.C slab of 75 mm thick over the duct.

Bed level of stream: 6.4 m below top of roadway. Assume suitable Hand rails &Kerb.

Draw 1) The cross section of the T beam and slab bridge. (25)

MODULE – III

V A plate girder for a bridge with 16m span has the following sections.

Web plate -1200 x8 mm. Flange angles-2 Nos back to back with 100x100x10mm

at top and bottom. Flange plates 2 Nos-330 x20mm on each flange.

Stiffeners -90x75x10mm.

Rivets-120 mm Φ at 150mm pitch.

Draw: 1) The cross section at centre of the plate girder (25)

OR

VI The details of a single laced compound column with welded connection are as given.

Steel stanchion– 2 Channels ISLC300 33.1-Kg/m placed back to back at 120mm clear spacing.

Tie plate –at top and bottom 320 x200x10mm on each side. Lacing-MS flat 100x6mm at 45°

Draw: 1) Sectional plan- (10)

2) Elevation (15)

MODULE-- IV

VII Draw to suitable scale-1) Longitudinal Section of canal drop with water cushion

U/S Bed level =100.00, FSL +101.5.50. Bank top level +103.00. Width of canal-7 m

Drop Wall: Top level +100..00, bottom Level =+96.00

Top width 1.m and bottom width 2.8m. Thick ness of solid apron 70cm,

Length of solid apron=6m and length of sloping grouted apron=4.8m with 40cm thickness.

Bed level of D/S =96.500 Top of D/S bank level ==99.500

Assume suitable abutments and wing walls: (25)

OR

VIII The hydraulic particulars of a Tank sluice with tower head are given below (25)

Sluice barrel: 0.60x0.75

Diameter of diaphragm stone: 0.45m

Thickness of RCC roof slab: 0.15m

Top width of tank bund: 2m

Side slope of tank bund: 2 : 1

Top bund level: +40.00

Maximum water level: +38.00

Full tank level: +37.500

Bottom level of sluice barrel: +34.500

Thickness of concrete foundation: 0.6

Internal diameter of tower head: 1.25m

Thickness of well steining: 0.45 m minimum, 0.65 m maximum

Size of stilling basin: 4.85 m x 2.5 m x. 1.5 m

Draw to a suitable scale, the longitudinal section of the sluice with the tower head.